

Solved Problems

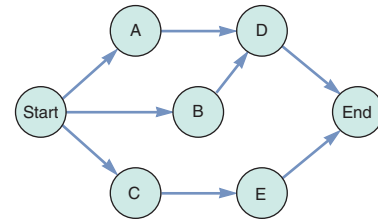
Virtual Office Hours help is available in [MyLab Operations Management](#).

SOLVED PROBLEM 3.1

Construct an AON network based on the following:

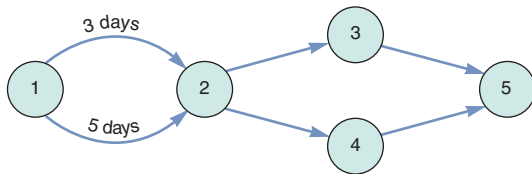
ACTIVITY	IMMEDIATE PREDECESSOR(S)
A	—
B	—
C	—
D	A, B
E	C

SOLUTION



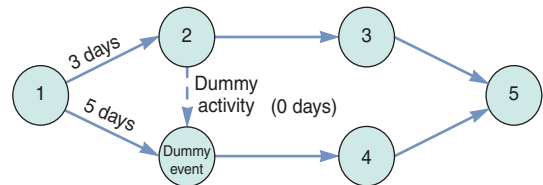
SOLVED PROBLEM 3.2

Insert a dummy activity and event to correct the following AOA network:



SOLUTION

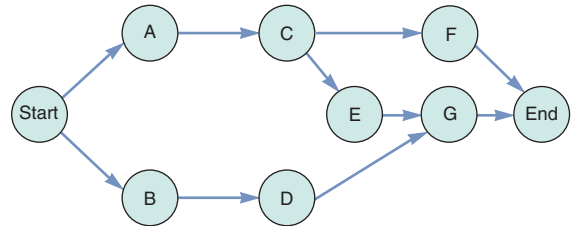
Because we cannot have two activities starting and ending at the same node, we add the following dummy activity and dummy event to obtain the correct AOA network:



SOLVED PROBLEM 3.3

Calculate the critical path, project completion time T , and project variance σ_p^2 , based on the following AON network information:

ACTIVITY	TIME	VARIANCE	ES	EF	LS	LF	SLACK
A	2	$\frac{2}{6}$	0	2	0	2	0
B	3	$\frac{2}{6}$	0	3	1	4	1
C	2	$\frac{4}{6}$	2	4	2	4	0
D	4	$\frac{4}{6}$	3	7	4	8	1
E	4	$\frac{2}{6}$	4	8	4	8	0
F	3	$\frac{1}{6}$	4	7	10	13	6
G	5	$\frac{1}{6}$	8	13	8	13	0



SOLUTION

We conclude that the critical path is Start–A–C–E–G–End:

$$\text{Total project time} = T = 2 + 2 + 4 + 5 = 13$$

and

$$\begin{aligned}\sigma_p^2 &= \sum \text{Variances on the critical path} \\ &= \frac{2}{6} + \frac{4}{6} + \frac{2}{6} + \frac{1}{6} = \frac{9}{6} = 1.5\end{aligned}$$

SOLVED PROBLEM 3.4

To complete the wing assembly for an experimental aircraft, Jim Gilbert has laid out the seven major activities involved. These activities have been labeled A through G in the following table, which also shows their estimated completion times (in weeks) and immediate predecessors. Determine the expected time and variance for each activity.

ACTIVITY	<i>a</i>	<i>m</i>	<i>b</i>	IMMEDIATE PREDECESSORS
A	1	2	3	—
B	2	3	4	—
C	4	5	6	A
D	8	9	10	B
E	2	5	8	C, D
F	4	5	6	D
G	1	2	3	E

SOLUTION

Expected times and variances can be computed using Equations (3–6) and (3–7) presented on page 110 in this chapter. The results are summarized in the following table:

ACTIVITY	EXPECTED TIME (IN WEEKS)	VARIANCE
A	2	$\frac{1}{9}$
B	3	$\frac{1}{9}$
C	5	$\frac{1}{9}$
D	9	$\frac{1}{9}$
E	5	1
F	5	$\frac{1}{9}$
G	2	$\frac{1}{9}$

SOLVED PROBLEM 3.5

Referring to Solved Problem 3.4, now Jim Gilbert would like to determine the critical path for the entire wing assembly project as well as the expected completion time for the total project. In addition, he would like to determine the earliest and latest start and finish times for all activities.

SOLUTION

The AON network for Gilbert's project is shown in Figure 3.17. Note that this project has multiple activities (A and B) with no immediate predecessors, and multiple activities (F and G) with no successors. Hence, in addition to a unique starting activity (Start), we have included a unique finishing activity (End) for the project.

Figure 3.17 shows the earliest and latest times for all activities. The results are also summarized in the following table:

ACTIVITY	ACTIVITY TIME				SLACK
	ES	EF	LS	LF	
A	0	2	5	7	5
B	0	3	0	3	0
C	2	7	7	12	5
D	3	12	3	12	0
E	12	17	12	17	0
F	12	17	14	19	2
G	17	19	17	19	0

Expected project length = 19 weeks

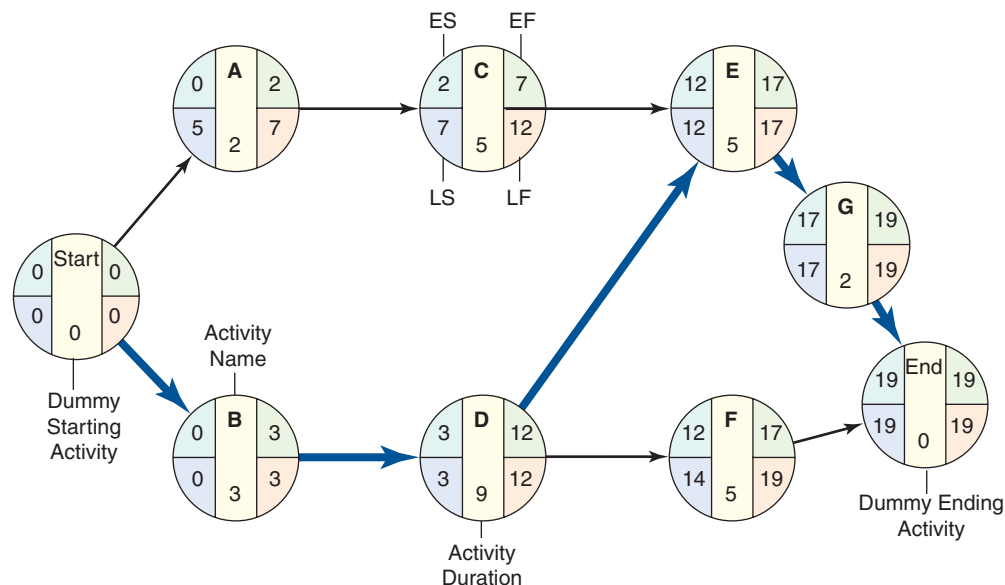
Variance of the critical path = 1.333

Standard deviation of the critical path = 1.155 weeks

The activities along the critical path are B, D, E, and G. These activities have zero slack as shown in the table.

Figure 3.17

Critical Path for Solved Problem 3.5



SOLVED PROBLEM 3.6

The following information has been computed from a project:

Expected total project time = $T = 62$ weeks

Project variance (σ_p^2) = 81

What is the probability that the project will be completed 18 weeks *before* its expected completion date?

SOLUTION

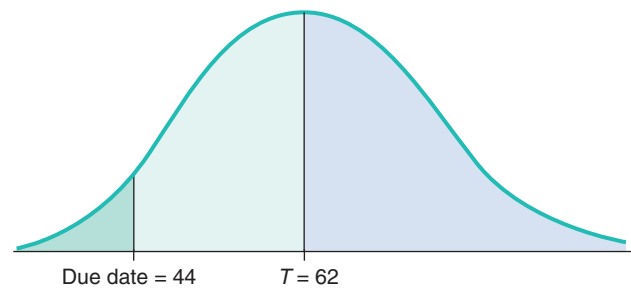
The desired completion date is 18 weeks before the expected completion date, 62 weeks. The desired completion date is 44 (or $62 - 18$) weeks:

$$\sigma_p = \sqrt{\text{Project variance}}$$

$$Z = \frac{\text{Due date} - \text{Expected completion date}}{\sigma_p}$$

$$= \frac{44 - 62}{9} = \frac{-18}{9} = -2.0$$

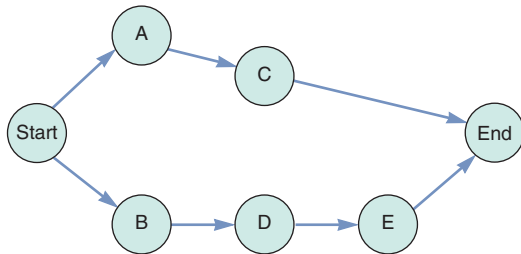
The normal curve appears as follows:



Because the normal curve is symmetrical and table values are calculated for positive values of Z , the area desired is equal to $1 - (\text{table value})$. For $Z = +2.0$ the area from the table is .97725. Thus, the area corresponding to a Z -value of -2.0 is .02275 (or $1 - .97725$). Hence, the probability of completing the project 18 weeks before the expected completion date is approximately .023, or 2.3%.

SOLVED PROBLEM 3.7

Determine the least cost of reducing the project completion date by 3 months based on the following information:



ACTIVITY	NORMAL TIME (MONTHS)	CRASH TIME (MONTHS)	NORMAL COST	CRASH COST
A	6	4	\$2,000	\$2,400
B	7	5	3,000	3,500
C	7	6	1,000	1,300
D	6	4	2,000	2,600
E	9	8	8,800	9,000

SOLUTION

The first step in this problem is to compute ES, EF, LS, LF, and slack for each activity.

ACTIVITY	ES	EF	LS	LF	SLACK
A	0	6	9	15	9
B	0	7	0	7	0
C	6	13	15	22	9
D	7	13	7	13	0
E	13	22	13	22	0

The critical path consists of activities B, D, and E.

Next, crash cost/month must be computed for each activity:

ACTIVITY	NORMAL TIME-CRASH TIME	CRASH COST-NORMAL COST	CRASH COST/MONTH	CRITICAL PATH?
A	2	\$400	\$200/month	No
B	2	500	250/month	Yes
C	1	300	300/month	No
D	2	600	300/month	Yes
E	1	200	200/month	Yes

Finally, we will select that activity on the critical path with the smallest crash cost/month. This is activity E. Thus, we can reduce the total project completion date by 1 month for an additional cost of \$200. We still need to reduce the project completion date by 2 more months. This reduction can be achieved at least cost along the critical path by reducing activity B by 2 months for an additional cost of \$500. Neither reduction has an effect on noncritical activities. This solution is summarized in the following table:

ACTIVITY	MONTHS REDUCED	COST
E	1	\$200
B	2	500
		Total: \$700